



Name: \_\_\_\_\_ Cohort/Batch: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

## AI CODING PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Emerging Technologies

TOTAL: 10 QUESTIONS

**Q1** What is AI Coding and what problem in Emerging Technologies does it solve?

Threat Model

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**Q6** How does AI Coding handle reliability and high availability?

Observability

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**Q2** What are the core components or concepts in AI Coding?

Architecture

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**Q7** What observability does AI Coding provide out of the box?

Lifecycle

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**Q3** How does AI Coding compare to its main alternatives?

Configuration

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**Q8** What are common performance bottlenecks in AI Coding?

Compliance

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**Q4** How do you get started with AI Coding for a new project?

Hardening

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**Q9** What security best practices apply when running AI Coding?

Testing

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**Q5** What configuration options most affect AI Coding's behavior in production?

SSO / IdP

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**Q10** What mistakes do teams commonly make when adopting AI Coding?

Tuning

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# ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

## A1 AI Coding is a tool or

Mitigation

AI Coding is a tool or platform that automates or simplifies a specific concern in the Emerging Technologies domain, reducing manual effort and operational complexity for engineering teams.

## A6 AI Coding provides HA through leader

Observability

AI Coding provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

## A2 AI Coding has key building blocks

Primitives

AI Coding has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

## A7 AI Coding exposes metrics (Prometheus)

Automation

AI Coding exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

## A3 AI Coding trades off simplicity against

Hardening

AI Coding trades off simplicity against feature richness differently than its alternatives. Choose AI Coding when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

## A8 Performance bottlenecks typically stem from

Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

## A4 Install AI Coding via its package

Gotchas

Install AI Coding via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

## A9 Run AI Coding with the principle

Assessment

Run AI Coding with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

## A5 Production-critical settings typically include

Federation

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

## A10 Common pitfalls include skipping the

Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.